Airflow and Thermal Comfort of

Contralateral Wall-Attached Jet Ventilation

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CFD Project Report

This report summarizes a computational fluid dynamics (CFD) study carried out at Chongqing University in 2024. The full paper is in Simplified Chinese.

Abstract

Wall-attached jet ventilation supplies air through a slot near the top of a wall. The jet clings to the wall, turns at the floor, and spreads across it as a thin layer of cool air, an “air lake”, that feeds the occupied zone. Earlier studies examined supply from one wall only, which limits how far the air lake reaches in deep rooms. This study proposes supplying air from two opposite walls and evaluates the airflow and thermal comfort of this contralateral arrangement in a classroom under summer cooling. The CFD model, built in Airpak 3.0, was first validated against published measurements of single-wall supply and then run for 15 cases of supply temperature (18, 20, and 22 °C) and supply velocity (0.6 to 1.4 m/s). The two jets form air lakes on both sides of the room that meet at its center. Temperature stratification starts at about seated head height, with a vertical difference of about 1.3 to 1.5 °C. At 22 °C every velocity keeps the occupant within the PMV and PPD comfort limits, 20 °C does so up to 1.2 m/s, and 18 °C fails at every velocity. The age of air at the occupant falls from about 800 s at 0.6 m/s to about 330 s at 1.4 m/s and is almost independent of supply temperature.

1. Background

Mixing ventilation and displacement ventilation each have known limits: mixing systems condition the whole room volume, and displacement systems deliver air at low momentum that does not travel far. Wall-attached jet ventilation combines high supply momentum with low velocities in the occupied zone. Air leaves a slot just below the ceiling, a small gap from the wall. The air between the jet and the wall cannot be replenished quickly, so a low-pressure region forms and pulls the jet onto the wall (the Coanda effect). The jet runs down the wall, separates and turns near the floor, and reattaches as a floor-hugging air lake. Its flow therefore has a vertical attached zone, an impingement zone, and a horizontal air-lake zone [1–4].

Most studies of this mode supply air from one wall. In a deep room the air lake slows and warms before it reaches the far side, and raising the supply velocity to push it farther causes drafts. This study supplies air from two opposite walls so that each jet serves half the room, and asks whether the arrangement keeps the occupied zone comfortable, which supply conditions work best, and how fresh the air is at the occupant.

2. Methods

2.1 Room and model

The room measures 5.4 × 7.0 × 3.16 m, the size of the laboratory used by Yin and colleagues in their single-wall experiments [2–4]. Two slot inlets, 2.0 × 0.05 m each, sit under plenums (2.5 × 0.5 × 0.5 m) near the top of the two long walls, 0.066 m from the wall surface. With an aspect ratio above 20:1, each slot behaves as a two-dimensional jet. A 0.2 × 0.2 m exhaust is at the center of the ceiling. The room is set up as a classroom, represented by one seated occupant, one computer, and one desk at its center

(Figure 1, Table 1). One end wall represents a sunlit exterior wall with windows and is held at 35.5 °C to represent an extreme summer day; all other surfaces are adiabatic.

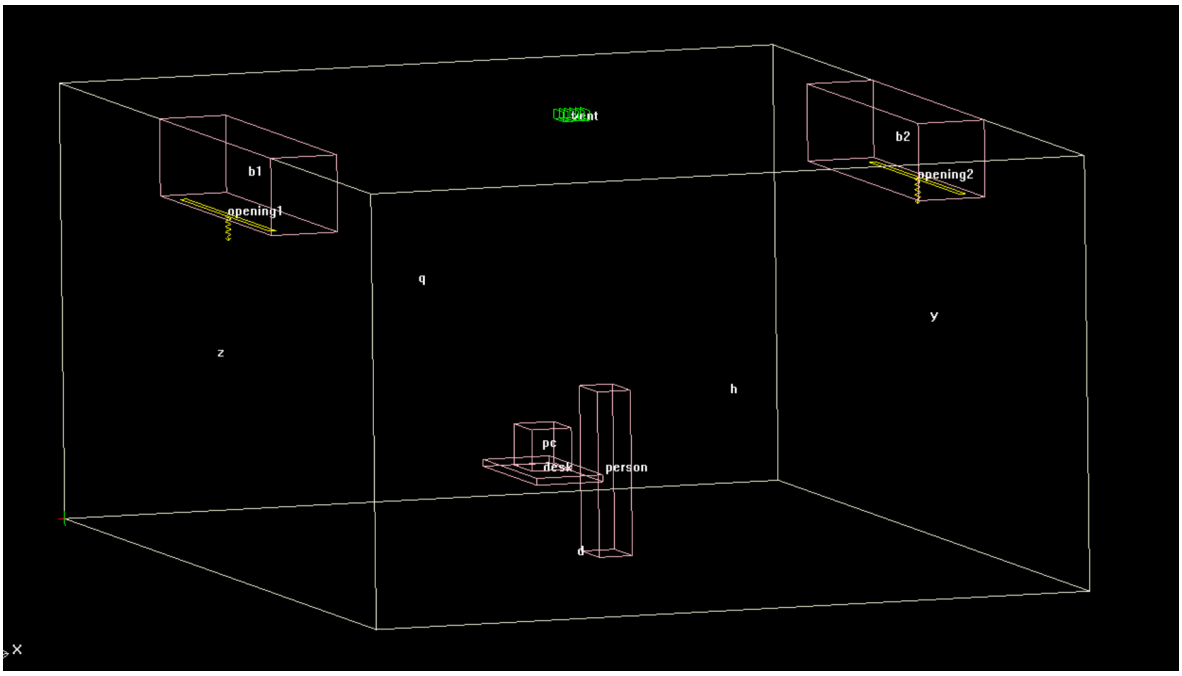


Figure 1. Airpak Model of the Room with the Two Supply Plenums (b1, b2), the Occupant, the Computer, and the Desk

Table 1. Objects and Boundary Conditions

Object	Size (m)	Boundary condition
Occupant	0.25 × 1.2 × 0.4	Heat source, 72 W
Computer	0.3 × 0.3 × 0.4	Heat source, 300 W
Desk	0.5 × 0.05 × 0.2	Adiabatic
Exterior wall	5.4 × 3.16	Fixed temperature, 35.5 °C
Supply slots (2)	2.0 × 0.05	Velocity inlet
Exhaust	0.2 × 0.2	Free outlet

2.2 Numerical setup

The model uses a structured hexahedral mesh of 556,347 cells (quality 0.98), refined around the plenums, slots, and exhaust. The flow is steady, incompressible, and turbulent, solved with the two-equation turbulence model available in Airpak 3.0, with gravity and buoyancy included (9.8 m/s² in the −y direction). The residual criterion is 10^{-4}.

2.3 Validation

Before the two-wall study, the single-wall laboratory case of Yin and colleagues was rebuilt with the same geometry, boundary conditions, and measurement positions, at a supply velocity of 2 m/s [2, 3]. The simulated centerline velocities in the vertical attached zone (0.16 to 2.45 m above the floor) and in the air-lake zone (0.35 to 3.0 m from the wall) followed the measured profiles and decay rates. The

largest deviations were at the two lowest points of the vertical zone and the two points nearest the wall in the air-lake zone, around the impingement region. The likely causes are differences in mesh and turbulence model: the original study used the SST k-ω model, which Airpak 3.0 does not offer, and two-equation k-ε models assume isotropic turbulence, which overpredicts turbulent kinetic energy where a jet impinges.

2.4 Cases and evaluation

Fifteen cases combine three supply temperatures (18, 20, and 22 °C) with five supply velocities (0.6, 0.8, 1.0, 1.2, and 1.4 m/s). Pre-runs showed that lower velocities do not keep the jet attached to the floor and higher ones cause drafts. Comfort is evaluated at a point 0.05 m in front of the seated occupant, at the height of the breathing zone (1.1 m). The study uses the predicted mean vote (PMV, comfortable between −0.5 and 0.5), the predicted percentage dissatisfied (PPD, at most 10%, Category B of ISO 7730) [6], and the age of air, the time air has spent in the room since entering it.

3. Results

3.1 Airflow

Each jet attaches to its wall within a short distance of the slot and runs down it with little mixing, so it reaches the occupied zone still cool and clean. At the floor the jet separates, turns, and reattaches as an air lake. Because both walls supply air, the two air lakes meet at the center of the room and form a second impingement zone, where the flow turns upward and mixes with the room air above the air lake (Figure 2). Where the air lake meets an obstacle, including the occupant, it climbs the surface, which helps remove heat and contaminants from around the body.

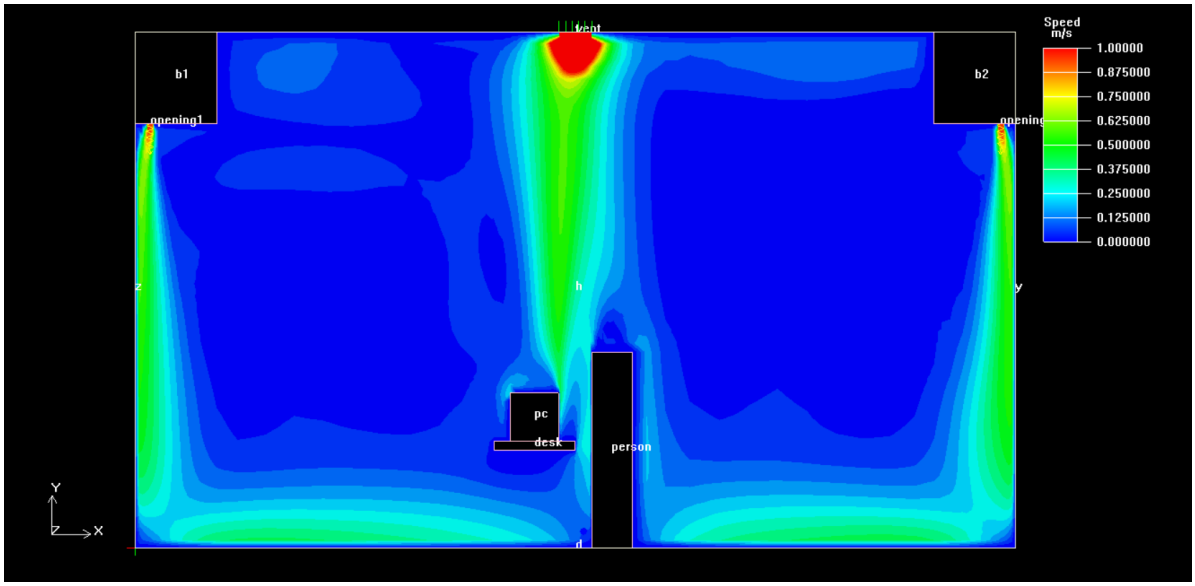


Figure 2. Example of the Velocity Field in a Vertical Section through the Room

In the vertical attached zone, the centerline velocity decays with distance from the slot. For the case of 22 °C and 0.8 m/s, a double-exponential function fits the decay with $R^2 = 0.987$:

$$\frac{u_m}{u_0} = A_1 exp(-\frac{x}{t_1}) + A_2 exp(-\frac{x}{t_2}) + y_0, \quad x = \frac{y*}{b}$$

where *u*m is the centerline velocity, *u*0 the supply velocity, *y** the distance below the slot, and *b* the slot width; the fitted values are A1 = 4.36, t1 = 1.26, A2 = −1.4 × 10^{-6}, t2 = −4.21, and y0 = 0.488. Earlier work on single-wall supply described the same decay with a power law, um/u0 = 1.12 (y*/b)^−0.21 [4].

3.2 Temperature

The supply forms a cool layer in the lower part of the room, with warmer air above (Figure 3). Stratification begins at about 1.1 m, roughly the head height of a seated person, so most of the body sits in stable, cool air. The vertical temperature difference between 0.1 m and 1.1 m at the occupant is between 1.25 and 1.45 °C in every case. Heat plumes rise from the computer and the occupant and carry warm air toward the ceiling exhaust. At a given supply temperature, the breathing-zone temperature falls as the supply velocity rises, because more cooling is delivered and the plume above the heat sources is diluted more strongly.

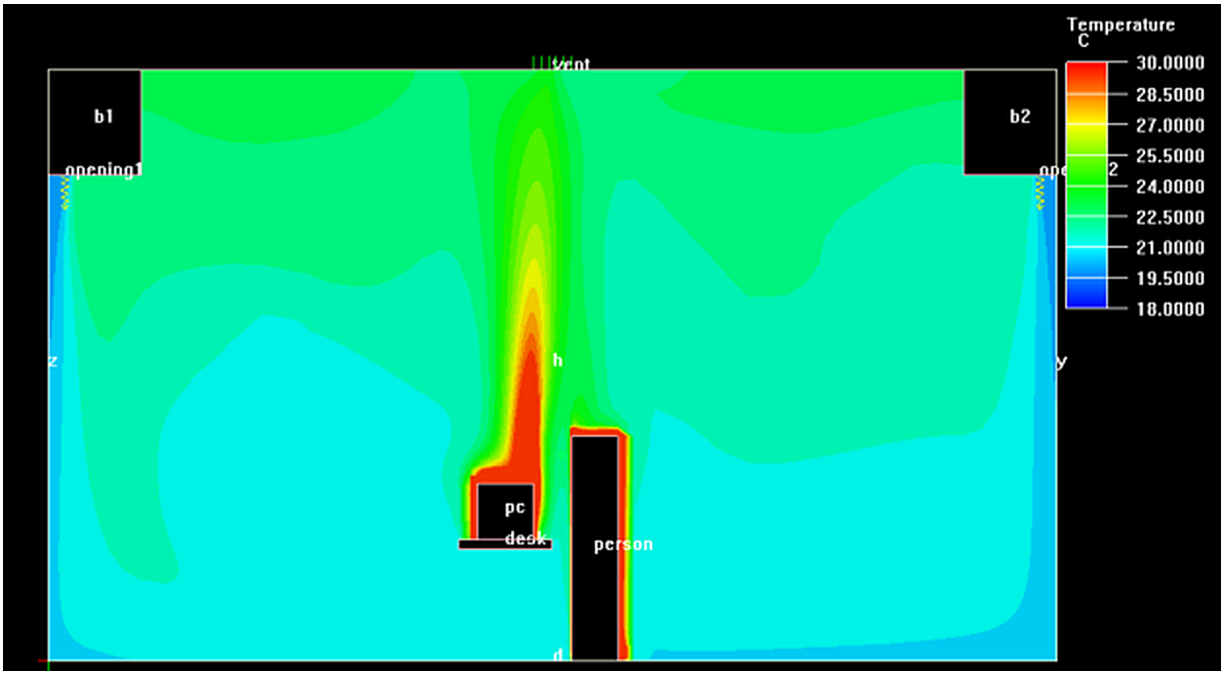


Figure 3. Temperature Field in a Vertical Section, Supply at 20 °C and 1.0 m/s

3.3 Thermal comfort

For the reference case of 20 °C and 1.0 m/s, PMV above the air lake is between −0.25 and 0.25 and is about −0.38 in the breathing zone. PPD is below 10% in most of the room; the exceptions are a strip about 0.2 m wide below each slot, part of the air lake, and the edges of the heat sources (Figure 4).

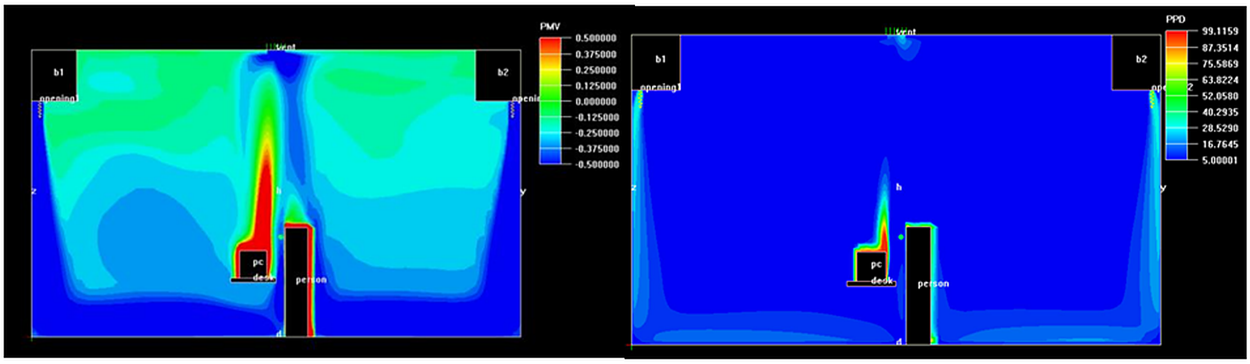


Figure 4. PMV (left) and PPD (right) in a Vertical Section, Supply at 20 °C and 1.0 m/s

Table 2 lists the results of all 15 cases at the occupant. Supply velocity drives comfort more than supply temperature does: at a fixed temperature, PMV falls steadily as velocity rises, and at a fixed velocity, each 2 °C step in supply temperature shifts PMV by only 0.27 to 0.34.

- **22 °C** keeps PMV between −0.18 and 0.10 and PPD between 5.0% and 5.6% at every velocity, the best of the three temperatures.
- **20 °C** meets both limits from 0.6 to 1.2 m/s. At 1.4 m/s it slightly exceeds them (PMV −0.51, PPD 10.5%).
- **18 °C** fails at every velocity: PPD is above 10% throughout, and PMV is below −0.5 at every velocity except 0.6 m/s, where it is −0.50.

Comfort is therefore achieved with a supply temperature of 20 to 22 °C, warmer than the supply air of a conventional cooling system. A warmer supply could reduce the chiller energy needed for the same comfort, but the study did not quantify this saving.

Table 2. Results at the Occupant for the 15 Cases

Supply (°C)	Supply (m/s)	Air temp. (°C)	Air speed (m/s)	Age of air (s)	PMV	PPD (%)	ΔT 0.1–1.1 m (°C)
18	0.6	21.80	0.21	808	−0.50	10.1	1.42
18	0.8	21.33	0.22	603	−0.59	12.4	1.45
18	1.0	20.96	0.24	502	−0.68	14.6	1.36
18	1.2	20.73	0.26	386	−0.74	16.7	1.33
18	1.4	20.30	0.29	320	−0.85	20.1	1.29
20	0.6	23.57	0.22	815	−0.22	6.0	1.41
20	0.8	22.99	0.22	586	−0.32	7.2	1.45
20	1.0	22.77	0.25	460	−0.38	8.0	1.29
20	1.2	22.44	0.26	372	−0.45	9.3	1.33
20	1.4	22.21	0.29	332	−0.51	10.5	1.25
22	0.6	25.48	0.21	835	0.10	5.2	1.42
22	0.8	25.01	0.23	638	0.01	5.0	1.35
22	1.0	24.47	0.25	458	−0.09	5.2	1.30
22	1.2	24.23	0.26	338	−0.14	5.4	1.30
22	1.4	24.13	0.29	332	−0.18	5.6	1.27

3.4 Age of air

In the well-ventilated parts of the room the age of air is about 490 s (Figure 5). Air above the air lake moves slowly and is older. The heat plume from the computer draws air past the occupant's face, so the breathing zone in front of the occupant has younger air than the area behind. At the occupant, the age of air falls from about 800 s at 0.6 m/s to about 330 s at 1.4 m/s, and the three supply temperatures give nearly identical values: air freshness is governed by supply velocity, not temperature.

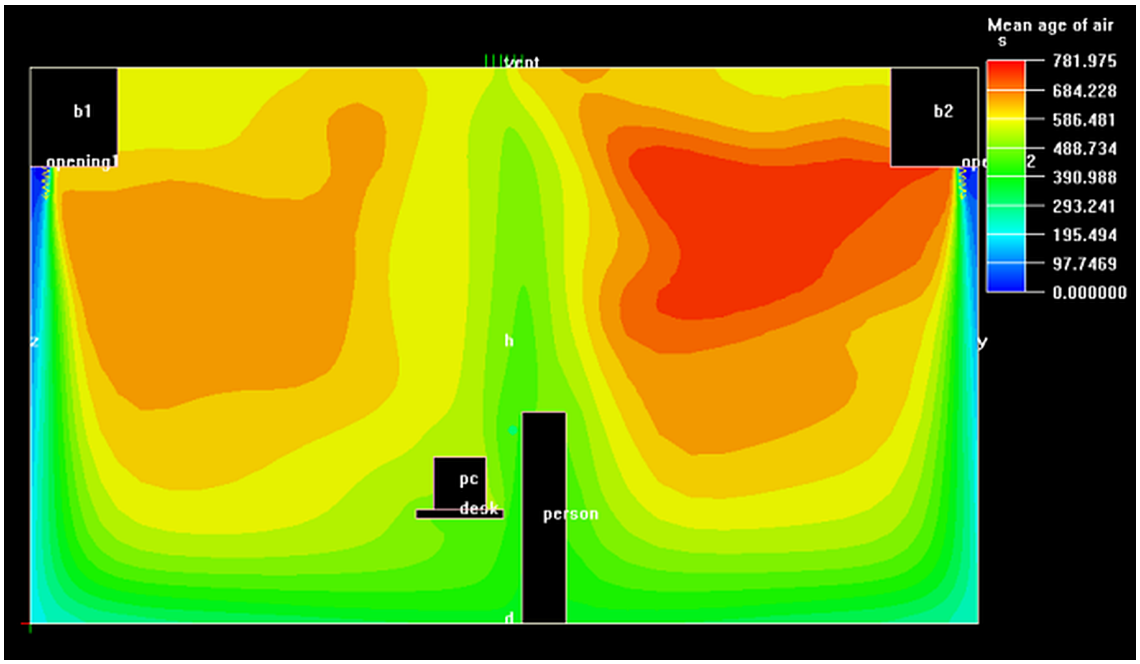


Figure 5. Age of Air in a Vertical Section, Supply at 20 °C and 1.0 m/s

4. Limitations

- **Simulation only.** No measurements were made of the two-wall arrangement. The model was validated against another team's measurements of single-wall supply, not of this configuration.
- **Turbulence model.** The two-equation model available in Airpak 3.0 is less accurate than the SST k-ω model in the impingement regions, where the validation showed the largest deviations.
- **One room and one probe point.** Comfort is reported for one occupant position in one room geometry, under steady summer conditions.
- **No energy calculation.** The energy argument is qualitative and rests on the higher supply temperature that still meets comfort limits.

5. Conclusion

Supplying air from two opposite walls produces air lakes on both sides of the room that keep the occupied zone within comfort limits, which addresses the reach limit of single-wall supply in larger rooms. Supply velocity has the larger effect on both comfort and air freshness, so design should set the

velocity first and then tune the supply temperature to the load. In the cases studied, a supply of 22 °C at 0.6 to 1.4 m/s, or 20 °C at up to 1.2 m/s, gives good comfort with about 1.3 to 1.5 °C of vertical stratification.

Data

The results of the 15 cases and the wall-jet velocity profile used for the decay fit are available at https://github.com/JiaweiQi3/wall-jet-ventilation-cfd.

References

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